⌈Def. Let G be a group, and $H \leq G$ be a subgroup,
Given an element $g \in G$, define left-coset of H as

$$gH \overset{def}{=} \{gh \mid h \in H\}.$$

⌈Lemma. Let G be a group, and $H \leq G$ be a subgroup. Then, for each $g_1, g_2 \in G$,

$$g_1 H = g_2 H \text{ if and only if } g_1^{-1} g_2 \in H.$$

Proof. Given $g_1, g_2 \in G$, suppose that $g_1 H = g_2 H$. Then, since $g_2 \in g_2 H = g_1 H$ $(1 \in H)$,

$$g_2 = g_1 h \text{ for some } h \in H, \text{ thus } g_1^{-1} g_2 \in H.$$

Conversely, let $g_1, g_2 \in G$ be given s.t. $g_1^{-1} g_2 \in H$. That is, $g_1 h = g_2$ for some $h \in H$.
Given $g_1 h_1 \in g_1 H$, $g_1 h_1 = g_2 h^{-1} h_1 \in g_2 H$. Given $g_2 h_2 \in g_2 H$, $g_2 h_2 = g_1 h h_2 \in g_1 H$ ☒

Q. In left coset sence, $g_1 H = g_2 H \iff g_1 g_2^{-1} \in H$ hold?
A. No! S_3, $H = \langle (1,2) \rangle$, $g_1 = (1\,3)$ and $g_2 = (1\,2\,3)$

⌈Thm. Lagrange Thm
If H is a subgroup of a group G, then $|G|/|H| = [G : H]$
Proof. Given a subgroup $H \leq G$, define a relation on G as;

$$x \sim y \iff xH = yH$$

Then, it is an equivalence relation, so a collection $\{gH \mid g \in G\}$ forms a partition.
Moreover, for each $g \in G$, a map

$$H \to gH \,;\, h \mapsto gh$$

is onto by definition and one to one by the cancelation law, so $|H| = |gH|$.